Iustin Surubaru

School of Mathematics, The University of Edinburgh
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Research Interests

Flat space holography, Celestial and Carrollian CFTs, mathematical structures of scattering amplitudes, twistor theory, twisted holography, on-shell methods and classical limit of scattering amplitudes

Education

Ph.D. in Mathematical Physics, The University of Edinburgh — 2023 - 2027
Title: Celestial holography and twistor theory
Advisor: Dr. Timothy Adamo

M.Sc. in Physics, The University of Oxford — 2019 - 2023
Award: Distinction

Publications

1. L. Cangemi, **I. Surubaru**, "Five-dimensional Geometry from Spinning Amplitudes", arxiv:2605.29831
2. T. Adamo, **I. Surubaru**, B. Zhu, "From AdS correlators to Carrollian amplitudes with the scattering equation", *JHEP* 02 (2026), arXiv:2512.03677
3. **I. Surubaru**, B. Zhu, "Carrollian amplitudes and holographic correlators in AdS_3/CFT_2", *Phys.Rev.D*, 112 (2025), arxiv:2504.07650
4. T. Adamo, **I. Surubaru**, "Twistorial chiral algebras in higher dimensions", *Class.Quant.Grav*, 42 (2025), arXiv:2501.09627
5. **I. Surubaru**, B. Zhu, "Conformal blocks from celestial graviton amplitudes", *JHEP* 06 (2025), arXiv:2501.05805

Invited Talks and Posters

- "MHV amplitudes in higher dimensions", PiTP workshop, Princeton — 2026
- "Five-dimensional geometry from amplitudes", Amplitudes conference, London — 2026
- "Recovering 5D black holes from amplitudes", NBMPS seminar, Durham — 2025
- "Five dimensional black holes from amplitudes", GRAMPA conference, Paris — 2025
- "Carrollian correlators and the flat space limit of AdS/CFT", Mathematical Physics group meetings, Edinburgh — 2025
- "Twistorial chiral algebras in higher dimensions", Celestial Holography Online Seminar, 2025
- "Koszul duality in celestial holography", GRIFT seminar, Edinburgh — 2024
- "Hodges' formula from twistor sigma models", STAMP seminar, Edinburgh — 2024

Conferences

- PiTP workshop, IAS, Princeton 2026
- “Amplitudes”, Queen Mary University, London 2026
- “String Math”, ICMS, Edinburgh 2026
- “Simons Collaboration on Celestial Holography annual meeting”, Simons Center, New York 2024 - 2026
- “AdS/CFT meets carrollian & celestial holography”, ICMS, Edinburgh 2025
- “GRAMPA”, Institut Henri Poincaré, Paris 2025
- “From Good Cuts to Celestial Holography”, Oxford 2025
- “From Asymptotic Symmetries to Flat Holography”, GGI, Florence 2025
- “Twistors in Geometry & Physics”, INI, Cambridge 2024
- “Amplitudes”, IAS, Princeton 2024

Awards and Prizes

- STFC PhD studentship 2023 - 2027
- Summer research studentship, Rudolf Peierls Centre for Theoretical Physics 2022
- New College Scholar, New College, Oxford 2020 - 2022
- Burden-Griffiths award, New College, Oxford 2020
- First prize, ITYM competition 2018
- Silver medal, romanian physics olympiad 2018 - 2019

Teaching Experience

Teaching Assistant, The University of Edinburgh 2024 - 2026
Fundamentals of Pure Mathematics, Linear Algebra, Classical Mechanics, Several Variable Calculus and Differential Equations
Private tutor, Leading Education 2023 - 2024
Helped international students with Oxford physics and mathematics exam preparation